

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Architected **distributed** platform services handling millions of daily transactions at **99%+ uptime** in a **large-scale** cloud computing environment.
- Engineered a pluggable **distributed executor** routing workloads across **Docker** and **AWS Lambda** with an **S3**-backed artifact store, designing for **fault tolerance** under scale-sensitive failure conditions.
- Owned full-stack delivery end-to-end, **React/TypeScript** frontend through **Python/FastAPI** backend, shipping in **agile** weekly cycles.
- Partnered with cross-functional model and platform teams to turn ambiguous, broadly defined problems into shipped production systems.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency 420ms to 140ms** across **50K+ daily requests** by redesigning synchronous serving into an **async, fault-tolerant** architecture on **GCP Cloud Run** with **autoscaling**.
- Improved **retrieval recall 61% to 89%** on a 2K-query eval set by rebuilding the **indexing** and query pipeline with **hybrid dense-sparse retrieval** and **re-ranking**.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Architected and operated **25+ production microservices** in **Java, C++**, and **SQL**-backed **distributed storage** systems for **1M+ users**, sustaining **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**.
- Standardized **failure-handling** contracts across 25+ financial services, reducing production escalations **60%** in one quarter via consistent **retry, timeout, and propagation semantics**.
- Redesigned a nightly batch data processing pipeline, cutting runtime from **8 minutes to 90 seconds**, and mentored junior engineers on **distributed-systems** design through full **SDLC**.

PROJECTS

Distributed KV Store / VPC Control Plane Simulator

github.com/ganeshhgupta/distributed-kv-store

- Raft-consensus** distributed storage engine handling concurrent writes with **linearizable reads, log compaction, and leader election**; companion VPC control plane simulator modeling subnet routing.

RootCause AI

github.com/ganeshhgupta/rootcause-ai

- Multi-agent** system automating production incident diagnosis via **tool-use** chains; routes alerts to specialized sub-agents and synthesizes root cause reports.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Full-stack eval and observability platform with **FastAPI** backend, **React/TypeScript** dashboard, and **PostgreSQL**, exposing **REST APIs** for scoring and latency tracking at scale.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Jul 2016 – Jul 2020

SKILLS

Languages: Java, C++, Python, TypeScript, SQL

Distributed Systems: Scalability, Fault Tolerance, Consensus Protocols, Indexing & Query Systems, Large-Scale Storage

System Design: Architecture, Agile/Full SDLC, Code & Design Review

Cloud & Data: AWS (EC2, RDS, SQS, S3, Lambda), GCP, Docker, Kubernetes, PostgreSQL, CI/CD

CERTIFICATIONS

AWS Certified Developer Associate | Google Cloud Professional Machine Learning Engineer